

Custom names of Interaction entities render at wrong location

MC-269996 Data Packs

Status	Resolved / Fixed
Affected	1.20.4, 24w13a, 1.20.6, 1.21
Fixed in	24w33a
Reported	2024-03-27

STEPS TO REPRODUCE

1. Launch Minecraft 1.20.3 or later.
2. Create a new Creative world with default settings.
3. Open the chat and execute `/summon interaction ~ ~ ~ {CustomName:"Test",CustomNameVisible:1,width:0.5,height:1.975}`.
4. Execute `/summon armor_stand ~ ~ ~ {CustomName:"Test",CustomNameVisible:1}`.
5. Execute `/summon armor_stand ~ ~ ~ {CustomName:"Test",CustomNameVisible:1,Marker:1}`.
6. Look at the summoned entities so their custom nameplates are visible.
7. Observe that the Interaction entity's nameplate renders as if the entity has zero height, similarly to the marker Armor Stand, instead of rendering above it at the specified height of 1.975 like the regular Armor Stand.

OBSERVED BEHAVIOUR

Actual behavior

The Interaction entity's custom nameplate is rendered at the entity's base, as though the entity's height were `0`, regardless of the `height` value specified in its NBT data. For example, an Interaction entity summoned with `height:1.975` displays its name at the same vertical position as a marker Armor Stand rather than above the top of its configured dimensions.

The Interaction entity's configured width and height are therefore not reflected in the dimensions used to calculate the custom nameplate's position. Its hitbox may have the specified dimensions, but the client continues to use zero for the entity's cached height when rendering the nameplate.

EXPECTED BEHAVIOUR

The Interaction entity's custom name should render above the entity at a vertical position determined by its specified height. With a height of 1.975, the nameplate should appear at approximately the same height as the nameplate of a regular Armor Stand, rather than at the entity's base as though it had zero height. Its custom-name visibility behavior should otherwise remain unchanged.

ENVIRONMENT

Minecraft Java Edition Versions: 1.20.4, 24w13a, 1.20.6, 1.21 (includes snapshots)

ORIGINAL DESCRIPTION

Since 1.20.3, Interaction entities have been able to display custom names given the resolution of MC-236501

Resolved

This is a great convenience when it comes to custom content creation for artificial mobs. It can allow with one entity, a hitbox and a naturally positioned and functional nametag (i.e. visible when looked-at or always respecting "CustomNameVisible"). However, currently to take full advantage of this functionality a modified server is required to force the game client understand the height (and width) of the desired Interaction entity dimensions.

As is, summoning an Interaction entity with a custom name shows the nameplate rendered as if the height of the entity was zero, instead of above what should be the real height of the entity. This is because the internally the cached `Entity#dimensions` field of the entity is not recomputed (as would by `Entity#refreshDimensions()`) when the width or height of Interaction entity changes, leading to `Entity#getBbHeight()` and `Entity#getBbWidth()` always being zero. This inconsistency also impacts the server, however those effects are less observable.

A modified server can workaround this oversight by forcing a dimension refresh by syncing an arbitrary Entity#DATA_POSE change when updating width and height, though being mindful to order the Pose last in the synched entries.

For data pack makers however, nothing can be done to fix the erroneous nameplate rendering, limiting the possibilities of the custom named Interaction entity usage.

Steps to reproduce:

- In Minecraft 1.20.3 or later, summon an Interaction entity with a custom name and width/height matching an Armor Stand using the command:

```
/summon interaction ~ ~ ~ {CustomName:"Test",CustomNameVisible:1,width:0.5,height:1.975}
```

- Summon a regular Armor Stand with custom name using the command:

```
/summon armor_stand ~ ~ ~ {CustomName:"Test",CustomNameVisible:1}
```

- Summon a marker Armor Stand with custom name using the command:

```
/summon armor_stand ~ ~ ~ {CustomName:"Test",CustomNameVisible:1,Marker:1}
```

Expected results:

- The Interaction entity's nameplate should render above the entity at a height corresponding to the specified height

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-269996>